Association of Residential Prices with Housing Features: Bedrooms, Bathrooms, Square Footage, Year Built, Floors and Condition

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1 Data and Research Question

Housing prices are influenced by many factors of a property. Features such as number of bedrooms, bathrooms, square footage, condition and more plays a role in the market value of a home.

Primary Goal: In this project, a housing dataset obtained from Kaggle containing 4600 residential properties is used to analyze how structural characteristics of houses relate to housing prices. The quantitative response variable is the residential price in USD.

The six predictors used in this model are number of **bedrooms**, **bathrooms**, **sqft_living** (square footage of home), **floors** (.5 denoting an attic space), **yr_built** (year home was built), and **condition** (rating scale 1 = poor – 5 = best)

Research Question: This study investigates the relationship between residential price and number of bedrooms, bathrooms, square footage, floors, year built, and condition.

2 Exploratory Data Analysis



Figure 1:
Scatterplots of Response vs. Key Continuous Quantitative Predictors

EDA cont.

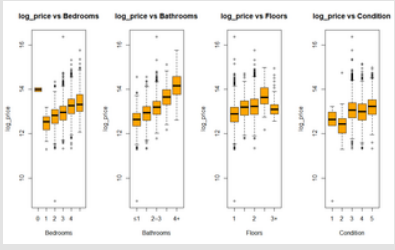


Figure 2:
Boxplots of Response vs. Key Categorical Quantitative Predictors

The response variable (**price**) was severely right skewed, so a **log transformation** is used throughout this regression analysis to improve model fit.

- Figure 1 suggests a positive association between square footage and log price.
- Figure 2 suggests a positive association between bedrooms, bathrooms, floors, condition, and log price.

EDA motivated inclusion of these predictors due to positive trends.

Basic Summary Statistics

- Average price of property = 551963 (dollars)
- Maximum price of property = 26590000 (dollars)
- Averages of bedrooms = 3.1, bathrooms = 2.2, floors = 1.5, sqft_living = 2139, yr_built = 1971, condition = 3.5

Selection Criteria: Forward selection with adjusted R^2 = 0.5105032 when using all 6 predictor variables in this initial model.

3 Model Specification

Based on EDA and adjusted R^2 selection criteria and forward selection methods, the initial model with most adjusted R^2 = 0.5105032 can be obtained:

$$log\,(price) = \beta_0 + \beta_1 bedrooms + \beta_2 bathrooms + \beta_3 sqft_living + \beta_4 floors + \beta_5 yr_built + \beta_6 condition$$

Model 1: Initial Model

- 51.05% proportion of variability is explained by the model after accounting for the 6 predictors used.
- This model also has the least AIC = 3922.305.

4 Estimation and Interpretation

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Coefficients:
                 Estimate Std. Error t value Pr(>|t|)
(Intercept)     2.056e+01  4.616e-01  44.531  < 2e-16
bedrooms       -8.602e-02  7.890e-03 -10.903  < 2e-16
bathrooms       1.217e-01  1.243e-02   9.795  < 2e-16
sqft_living     5.340e-04  1.758e-05  30.373  < 2e-16
I(sqft_living^2) -2.777e-08  2.267e-09 -12.247  < 2e-16
floors          1.459e-01  1.235e-02  11.820  < 2e-16
condition       5.155e-02  8.913e-03   5.784 7.79e-09
yr_built       -4.490e-03  2.333e-04 -19.246  < 2e-16
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Table 1: Summary of Model 2 (t-test)

All predictors are statistically significant as seen with the p-values being below α = 0.05, providing evidence of a relationship between housing prices and all 6 predictors. **Holding all other variables constant:**

- a one-unit increase in **sqft_living**^{2} is associated with an estimated decrease of 2.78×10^{-8} units in the expected value of log(price).
- a one-unit increase in **bathrooms** is associated with an estimated increase of 0.122 units in the expected value of log(price).

The adjusted R^2 is .5381, indicating that the **final model** explains 53.81% of the variability in the housing prices.

5 Model Comparison

Model diagnostic plots suggest linearity assumption violation. To treat this, a polynomial term was added for each predictor and their subsequent AIC was compared. At the lowest AIC = 3776.47545 the following model is obtained with a sqft_living polynomial term:

$$log\,(price) = \beta_0 + \beta_1 bedrooms + \beta_2 bathrooms + \beta_3 sqft_living + \beta_4 sqft_living^2 + \beta_5 floors + \beta_6 yr_built + \beta_7 condition$$

Model 2: Final Model Choice

From the tests above, we can see that this model is the better choice as the adjusted R^2 has increased from 51.05% to 53.81%. Models were compared using adjusted R^2 and AIC.

6 Assumptions and Diagnostics

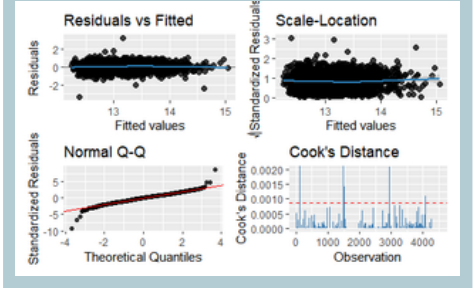


Figure 3: Diagnostic Panel

- Leverage values show no undue influence
- Before sqft_living^2: violations of **linearity**, **normality**, **constant variance**, and **outliers**
- Adding sqft_living^2 fixed **linearity**, **normality**, and **constant variance**
- A few points exceed Cook's D cutoff, but influence is small → all retained
- **Limitations**: The data are observational, the results describe associations rather than causal effects.